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Deep eutectic solvents used as the green media for the efficient extraction of caffeine from Chinese dark tea

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ABSTRACT

In recent years, more and more attention are paid to the green media in the extraction field. Deep eutectic solvents (DESs) are considered as excellent substitutes to traditional organic solvents and ionic liquids (ILs). In this work, DESs were used to extract caffeine from Chinese dark tea (CDT), then caffeine was recovered from the crude DES extract using macroporous resins. The factors affecting the solid-liquid extraction, including DES concentration, liquid-solid ratio, extraction time, extraction temperature, and system pH were studied in the single factor experiments. The Kamlet-Taft polarity parameters of DESs were determined to analyze the proper hydrogen-bond interaction between DESs and caffeine. Then the major factors of DES concentration, liquid-solid ratio, and extraction time were optimized by response surface methodology (RSM). The results showed that the maximum extraction yield of 26.78 mg/g were obtained using the DES choline chloride-lactic acid (ChCl-LA) as extractant under the optimized conditions. Moreover, the kinetics model of caffeine extraction was established using the Fick's Second Law, the experimental results were well fitted by this model. This extraction of caffeine from CDT using DESs is much simpler and environmentally friendly with low cost and high extraction efficiency.

1. Introduction

Caffeine (1, 3, 7-trimethylxanthine), one of the most popular alkaloid substance in the world, has been studied over the latest few decades due to its stimulant effects on the nervous, muscular and cardiovascular systems [1]. Caffeine can cause smooth muscle relaxation, reduce physical fatigue, and improve athletic performance in some cases [2,3], so it was used in a wide array of pharmaceutical products and beverages [4]. Caffeine can be commonly found in coffee seeds, tea leaves, cocoa beans as a natural compound [5]. Although caffeine plays a positive role in some cases, excess intake will also be adverse to the human health [6]. Recently researchers show increasing interest in caffeine extracted from tea. Tea (*Camellia Sinensis* L.) is a traditional Chinese drink and has become an important agricultural product. Traditionally, tea is mainly divided into dark tea, green tea, oolong tea, white tea, black tea, and yellow tea. Chinese dark tea (CDT) is a microbial post-fermented tea mainly distributed in Southwestern China, including Pu-erh tea (Yunnan), Ya'an Tibet tea (Sichuan), Qingzhuan tea (Hubei), Liubao tea (Guangxi), Fu-brick tea (Hunan), and so on. [7]. Fu-brick tea is mainly planted and processed in Anhua County, Yiyang City, Hunan Province, and which becomes an important industry in this area. The traditional extraction methods such as heat reflux extraction or Soxhlet extraction using organic solvents were the main methods to obtain caffeine from tea [8]. However, these methods have some disadvantages such as time consuming, low extraction yield and potential environmental pollution. The extraction of caffeine from tea using supercritical carbon dioxide is green and effective [9], but sophisticated equipment is needed and the cost is much higher. Therefore, to explore a green and simple method for the extraction of caffeine is of great significance.

Deep eutectic solvents (DESs) were first introduced by Abbott and his coworkers [10], which were prepared from hydrogen bond acceptors (HBAs) and hydrogen bond donors (HBDs). DESs possess similar properties to ILs, including nonflammability, high thermal stability, low vapor pressure, and high dissolving capacity [11,12]. However, DESs have some distinct advantages compared to conventional ILs, such as good biodegradability, low toxicity, easy preparation, and low cost [13–15]. Based on aforementioned properties, DESs were widely used in the fields of functional material preparation [16], friction lubrication [17], electrochemistry [18], gas separation [19], and extraction of bioactive compounds [20]. In the extraction field, DESs were usually used as extraction media in the solid-liquid extraction [20], aqueous

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two-phase extraction [21], magnetic solid-phase extraction [22], solid phase microextraction [23], dispersive liquid-liquid microextraction [24], and dispersive liquid-phase microextraction [25]. In recent years, DESs were developed as green media for the solid-liquid extraction of polyprenyl acetates from *Ginkgo biloba* leaves [26], ginseng saponins from white ginseng [27], flavonoids from sea buckthorn leaves [28], polysaccharides from *Dioscorea opposita* Thunb [29], and phenolic metabolites from *Carthamus tinctorius* L. [20].

In this study, eight DESs were chosen as the green media for the extraction of caffeine from CDT. Some waste or poor quality CDT leaves produced in the production process were chosen as the sources of raw materials. The DES ChCl-LA was screened as the optimal extractant. The major factors affecting the extraction were optimized by RSM to obtain the optimal extraction conditions. The Kamlet-Taft polarity parameters of DESs and the extraction kinetics were studied. Moreover, caffeine was recovered from the crude DES extract using macroporous resins.

2. Materials and methods

2.1. Plant materials and reagents

The CDT Fu-brick tea was provided by Yiyang Longling Dark Tea Co., Ltd, Yiyang City, China. The leaves were placed in the air dry oven at 40 °C for 12 h till no weight loss, then smashed to 60 meshes particle size. The caffeine standard (HPLC purity of > 98%) was purchased from Alladin Reagent Co., Ltd. (Shanghai, China). Choline chloride (> 98%), D-sorbitol (> 98%), urea (> 99.5%), oxalic acid(> 98%), citric acid (> 98%), L-(+)-diethyl tartrate (> 98%), zinc chloride (> 98%), L(+)-lactic acid (> 90%), and glycerol (> 98%) were purchased from Shanghai Titan Technology Co. Ltd (Shanghai, China). The dyes of 2,6-dichloro-4-(2,4,6-triphenyl-N-pyridino)-phenolate (Reichardt's betaine dye 33, RD), N, N-diethyl-4-nitroaniline (DENA), and 4-nitroaniline (NA) were provided by Molbase Biotechnology Co., Ltd. (Shanghai, China). The macroporous resins (Model DM-130, AB-8, D101, H103, NKA-2, XAD-2 XDA-200, and X-17) were purchased Jiangsu Color Kesi Si Resin Co., Ltd. (Jiangsu, China). The physical properties of all the macroporous resins used in this study were shown in Table S1. HPLC grade methanol used as mobile phases was purchased from TEDIA Company, Inc. (Fairfield, OH, USA).

2.2. Preparation of DESs

The preparation of DESs was conducted according to the heating method reported in the previous literatures [30,31]. DESs used in this study were prepared by combining the components in certain molar ratios and then stirring the mixture at 80 °C till a homogeneous liquid formed. The materials and molar ratios of these DESs were shown in Table 1.

2.3. Extraction of caffeine from CDT using different DESs

To a 10 mL tube, a certain amount of DESs, water, and 0.2 g CDT powder were added. The mixture was well shook by vortex and then placed in a water bath (set at different temperatures) for the ultrasound assisted extraction in a certain time. The tube was centrifuged at 10, 000 rpm for 5 min after extraction. The supernatant was withdrawn and diluted, then filtered using millipore filter (0.45 μm) before injection into the HPLC system. The extraction yield of caffeine was calculated by the Eq. (1):

$$\text{Extraction yield (mg/g)} = \frac{\text{Mass of caffeine determined in DES extract (mg)}}{\text{Mass of CDT leaves powder(g)}} \quad (1)$$

2.4. Experimental design using Box-Behnken design (BBD)

The BBD was used to investigate the four main factors affecting the caffeine extraction, including (A) extraction temperature, (B) extraction time, (C) liquid-solid ratio, and (D) DES concentration. The four-factor and three-level design were performed and the 29-runs experiments were done using the extraction yield of caffeine as the response value. Response surface design was studied to optimize the caffeine extraction conditions and predict the extraction yield.

2.5. Recovery of caffeine using macroporous resins

The solid-liquid extraction using DESs can obtain the crude extract, then eight kinds of macroporous resins (Seeing Table S1) were used to recover and enrich the caffeine by static adsorption and desorption. The detailed procedures were as follow: To a 150-mL Erlenmeyer flask, 1.0 g (dry weight) of macroporous resins were added. An additional 40 mL of crude extract was added and adsorbed for 6 h in a 25 °C constant temperature shaker at a speed of 100 rpm. When the adsorption reached equilibrium, the resins were separated from the solution by adsorption filtration. Then, 40 mL absolute ethanol was added to the above-mentioned resins separated from the solution by suction filtration. Desorption was performed with ultrasonic assist for 6 h in a 25 °C constant temperature shaker at a speed of 100 rpm. After accomplishing the desorption, the resins were separated from the solution by suction filtration. The adsorption capacity(Q_e, mg/g) and desorption ratio (D, %) of the macroporous resins were calculated by the Eqs. (2) and (3), respectively.

$$Q_e = \frac{V \times (C_a - C_b)}{M} \quad (2)$$

where V represents the volume (mL) of the sample solution; C_a represents the initial concentration (mg/L); C_b represents equilibrium concentration (mg/mL); M represents dry weight (g) of resin.

$$D = \frac{V_i \times C_i}{V \times (C_a - C_b)} \quad (3)$$

where V_i is the volume (mL) of desorption solution; C_i is caffeine concentration in desorption solution (mg/mL), V, C_a and C_b are the same as that in Eq. (2).

Table 1
The Kamlet-Taft polarity parameters for the 70 wt% DESs aqueous solution.

No.	Hydrogen bond acceptor	Hydrogen bond donor	Molar ratio	E_T (33)	Π^*	α	β	$\alpha \times \beta$
DES-1	choline chloride	D-sorbitol	1:1	66.804	1.244	2.216	0.493	1.093
DES-2	choline chloride	urea	1:2	67.355	1.227	2.253	1.238	2.789
DES-3	choline chloride	oxalic acid	1:1	66.185	1.288	2.173	0.825	1.792
DES-4	choline chloride	citric acid	1:1	67.433	1.373	2.248	0.714	1.604
DES-5	choline chloride	L-(+)-diethyl tartrate	1:1	65.880	1.770	2.118	0.840	1.779
DES-6	choline chloride	zinc chloride	1:1	56.618	1.406	1.543	1.253	1.737
DES-7	choline chloride	lactic acid	1:1	67.593	1.305	2.263	1.025	3.320
DES-8	choline chloride	propanetriol	1:2	66.185	1.253	2.175	0.599	1.302

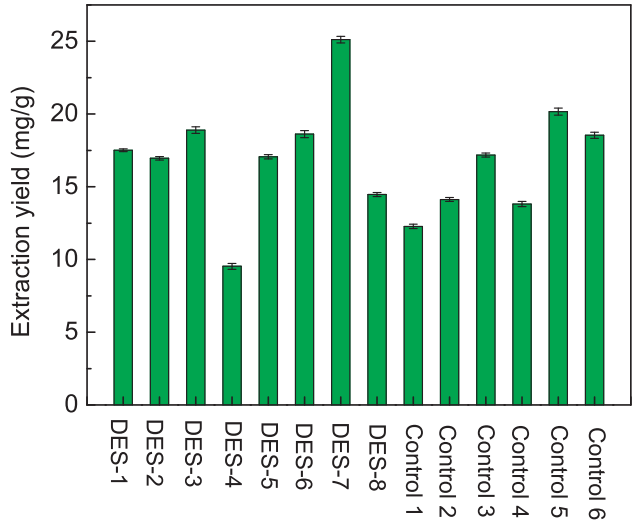


Fig. 1. The extraction yields of caffeine using various DESs. The extraction conditions were as follow: 0.2 g CDT leaves powder, 25:1 liquid-solid ratio, and 70 wt% DES concentration, 30 min extraction time, and 30 °C extraction temperature for each extraction. Control 1: 70 wt% choline chloride as extractant; Control 2: 70 wt% lactic acid as extractant; Control 3: a solution of simple mixing of choline chloride and lactic acid at a molar ratio of 1:1 and total mass fraction of 70%; Control 4: 100% water; Control 5: 70 wt% ethanol; and Control 6: 70 wt% methanol.

2.6. Kamlet-Taft polarity parameters

The hydrogen bond donating ability (α, hydrogen-bond acidity), hydrogen bond accepting ability (β, hydrogen-bond basicity), and dipole/polarizability (π^*) for ILs and DESs were widely quantified by the Kamlet-Taft parameters. The dyes with various concentrations (DENA and NA of 5.0×10^{-5} mol/L, RD of $2.2 \times 10^{-4} - 8.6 \times 10^{-3}$ mol/L) were mixed with DESs. The maximum absorption wavelength of the dye was measured using an UV-Vis spectrophotometer (Model UV-3000, Shanghai MAPADA Instruments Co., Ltd., Shanghai, China). The detailed procedures were as follow: the dyes were separately added to the 70 wt% DESs aqueous solution, then poured into a quartz cuvette, and subjected to wavelength scanning using the UV-Vis spectrophotometer. The maximum absorption wavelength was selected between 400 and 600 nm. When the maximum absorption wavelength was out of the scope of 400–600 nm, the first derivative of the spectrum was selected to obtain this value. The Kamlet-Taft polarity parameters were obtained from the Eqs. (4)–(8) [32]. Where ν_{max} is the maximum experimental wave number, E_T (33) is the solvatochromic parameter and λ_{max} is the maximum absorption wavelength.

$$\nu_{max}(cm^{-1}) = 10000/\lambda(nm) \quad (4)$$

$$E_T(33) = 28592/\lambda_{max}(\mathrm{RD}) \quad (5)$$

$$\pi^* = 0.314 \times [27.52 - \nu_{max}(\mathrm{DENA})] \quad (6)$$

$$\alpha = 0.0649 \times E_T(33) - 2.03 - 0.72\pi^* \quad (7)$$

$$\beta = [1.035 \times \nu_{max}(\mathrm{DENA}) + 2.64 - \nu_{max}(\mathrm{NA})]/2.8 \quad (8)$$

2.7. Establishment of the kinetic model for caffeine extraction

To study the mass transfer kinetics during the extraction of caffeine at different temperatures, the kinetics of solid-liquid extraction of caffeine was studied according to Fick's Second Law of diffusion. According to the plate model, it was assumed that the caffeine in the particles was evenly distributed, the diffusion coefficient of caffeine was unchanged, and no significant temperature gradient was established for caffeine and solvent. The formula for fitting the parameters is shown in Eq (9) [33]:

$$\ln[C_\infty/(C_\infty - C)] = k \times t + A \quad (9)$$

where C_∞ is the equilibrium extract concentration (mg/mL), C is the extract concentration at time t (mg/mL), k is the pseudo first-order kinetics constant, and A is an integration constant.

2.8. HPLC analysis

The caffeine samples were analyzed using an iChrom W5100 HPLC system (Dalian Elite Analytical Instruments Co. Ltd., Dalian, China). The Syncronis C_{18} column (4.5 × 250 mm, 5 μm) was used to analyze the samples. The conditions for analysis of the samples were as follows: mobile phase (methanol: 1% acetic acid aqueous solution = 30:70); flow rate: 0.8 mL/min with isocratic elution; detection wavelength: 274 nm; column temperature: 40 °C; sample volume: 10 μL. The caffeine standard with different concentrations (0.1, 0.5, 1.0, 1.5, 2.0 mg/mL) was injected into the HPLC system to calculate the calibration curve. The calibration curve was fitted by the software Origin 8.5, the result is shown in Eq (10):

$$Y = 83362X - 1806 (R^2 = 0.99998) \quad (10)$$

where Y is the peak area and X is the caffeine concentration. The standard solutions of caffeine were diluted from 0.1 to 2.0 mg/mL.

3. Results and discussion

3.1. Choice of the optimal DES

The screening of extractant used for the extraction of caffeine from CDT is the foremost work should be done. Eight commonly used DESs (Table 1) with 70 wt% concentration were considered as the extractants and six controls were designed, Control 1: 70 wt% choline chloride as extractant; Control 2: 70 wt% lactic acid as extractant; Control 3: a solution of simple mixing of choline chloride and lactic acid at a molar ratio of 1:1 and total mass fraction of 70%; Control 4: 100% water; Control 5: 70 wt% ethanol; and Control 6: 70 wt% methanol. As shown in Fig. 1, the maximum extraction yield was obtained using ChCl-LA as the extractant. This can be explained by the assumption that the stronger hydrogen bonding interaction can be occurred between caffeine and ChCl-LA. ChCl-LA used as extractant had larger extraction yield than the Controls 1–3, which indicated that the acting force between the DES and caffeine played a dominant role. The using of ChCl-LA as extractant resulted in the higher extraction yield than Controls 4–6, indicating this DES is more efficient than the common water and organic solvents in the extraction of caffeine from CDT.

To further confirm the assumption of hydrogen bonding interaction, the Kamlet-Taft polarity parameters were measured for 70 wt% DESs aqueous solution. The results for the α, β, and π^* values were shown in Table 1. The parameter π^* can provide a measure of the solvent polarizability [34], and the π^* values varied little and were in the range of 1.2–1.7 for all the tested DESs solution in this study. All the α values were > 1.0 and higher than most of the organic solvents as reported in the previous study [34]. Majority of the α values were around 2.2, and it proved that the parameter α was mainly driven by the HBA [34]. In this study, the same HBA of choline chloride was chosen, so the α values varied little. It was reported that the $\alpha \times \beta$ value can represent the hydrogen bond ability of DESs [35]. As to the DES ChCl-LA, a π^* value of 1.305 and the largest $\alpha \times \beta$ value were obtained, which indicated that maybe the hydrogen bond interaction between ChCl-LA and caffeine was the foremost factor influencing the extraction. As a matter of fact, it was also reported that there are many other factors influencing the extractability of DESs, including hydrophobic interactions, electrostatic interactions, and van der Waals forces [36]. Therefore, the extraction using DESs as extractants is a very complicated process.

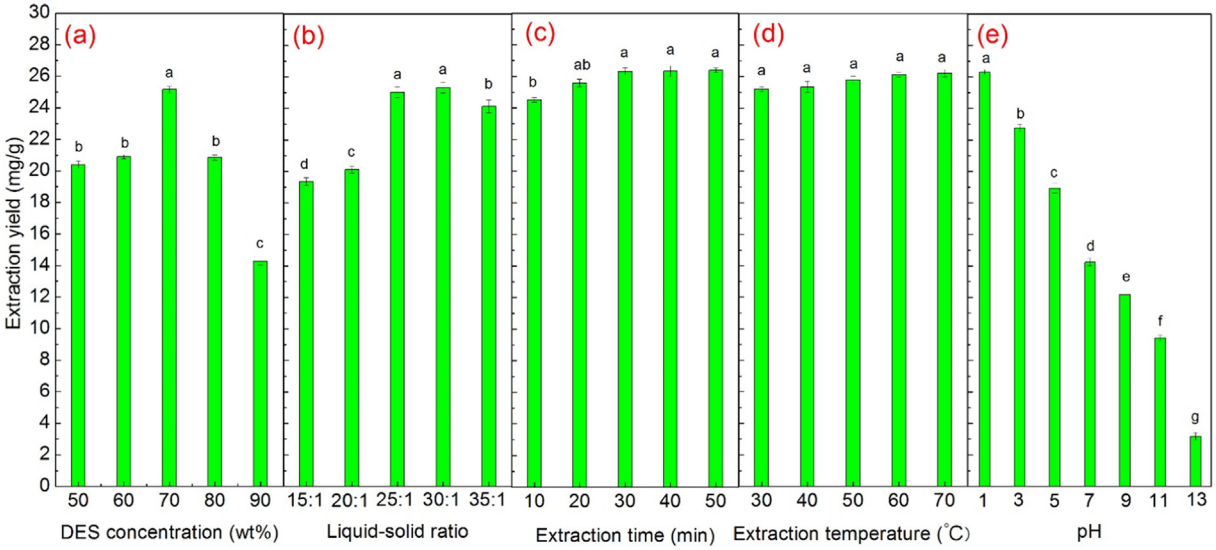


Fig. 2. Effect of single factor to the extraction yield of caffeine. **(a)** DES concentration, the extraction conditions were as follow: 25:1 liquid-solid ratio, 20 min extraction time, room temperature extraction, and pH was not adjusted. **(b)** liquid–solid ratio, the extraction conditions were as follow: 70 wt% DES concentration, 20 min extraction time, room temperature extraction, and pH was not adjusted. **(c)** extraction time, the extraction conditions were as follow: 70 wt% DES concentration, 30:1 liquid-solid ratio, room temperature extraction, and pH was not adjusted. **(d)** extraction temperature, 70 wt% DES concentration, 30:1 liquid–solid ratio, 30 min extraction time, and pH was not adjusted. **(e)** system pH, 70 wt% DES concentration, 30:1 liquid-solid ratio, 30 min extraction time, 60 °C extraction temperature. The Duncan's multiple range test was used for the statistical analysis. Different letters in the same series indicate significant difference at $p < 0.05$ level.

Table 2
Analysis of variance for the regression model.

Source	Sum of Squares	Degree of freedom	Mean square	*F* value	*P* value
Model	35.02	14	2.5	542.84	< 0.0001
A	0.14	1	0.14	31.3	< 0.0001
B	0.024	1	0.024	5.28	0.0375
C	6.25	1	6.25	1355.61	< 0.0001
D	0.6	1	0.6	130.83	< 0.0001
AB	1.32×10^{-6}	1	1.32×10^{-6}	2.87×10^{-4}	0.9867
AC	0.029	1	0.029	6.29	0.0251
AD	2.45×10^{-3}	1	2.45×10^{-3}	0.53	0.4784
BC	9.33×10^{-3}	1	9.33×10^{-3}	2.02	0.1766
BD	6.40×10^{-4}	1	6.40×10^{-4}	0.14	0.715
CD	9.44×10^{-3}	1	9.44×10^{-3}	2.05	0.1743
A^2	0.56	1	0.56	122.29	< 0.0001
B^2	0.19	1	0.19	41.64	< 0.0001
C^2	26.47	1	26.47	5744.38	< 0.0001
D^2	3.63	1	3.63	787.53	< 0.0001
Residual	0.065	14	4.61×10^{-3}		
Lack of Fit	0.039	10	3.89×10^{-3}	0.61	0.7617
Pure Error	0.026	4	6.40×10^{-3}		
Cor Total	35.09	28			

3.2. Single factor experiments for the solid-liquid extraction of caffeine from CDT

3.2.1. Effect of DESs concentration

The DES ChCl-LA was chosen as the optimal extractant, and its concentration should be optimized. The effect of the DES concentration of 50–90 wt% was investigated. The results were shown in Fig. 2(a), it can be seen that the extraction yields increased with the increasing of ChCl-LA concentration at 50–70 wt%, however, the further increasing in DES concentration resulted in the decreasing of extraction yields. The increase of water in DES can decrease the viscosity of the extraction system, which is beneficial for the extraction. However, the polarity simultaneously increased with increase of water content [34], which can destroy the hydrogen-bond interaction between DES and caffeine and make this DES solution being not suitable for the extraction. It was also well reported by previous studies that the addition of 25–30 wt% water can enhance the extraction efficiency, however, when the water content was > 50 wt%, the structure of DESs will be affected [37,38]. Therefore, 70 wt% ChCl-LA concentration was chosen for the further studies.

3.2.2. Effect of liquid-solid ratio

The effect of liquid-solid ratio of 15:1, 20:1, 25:1, 30:1 and 35:1 was investigated in this work. As shown in Fig. 2(b) that the extraction yields increased with the increasing of liquid-solid ratio from 15:1 to 30:1, and then decreased with further increase of liquid amount. The other studies had also reported that too low or too high liquid-solid ratio will reduce the amount of extraction or increase the cost as the amount of solution increases [39,40]. Therefore, the liquid-solid ratio of 30:1 was selected.

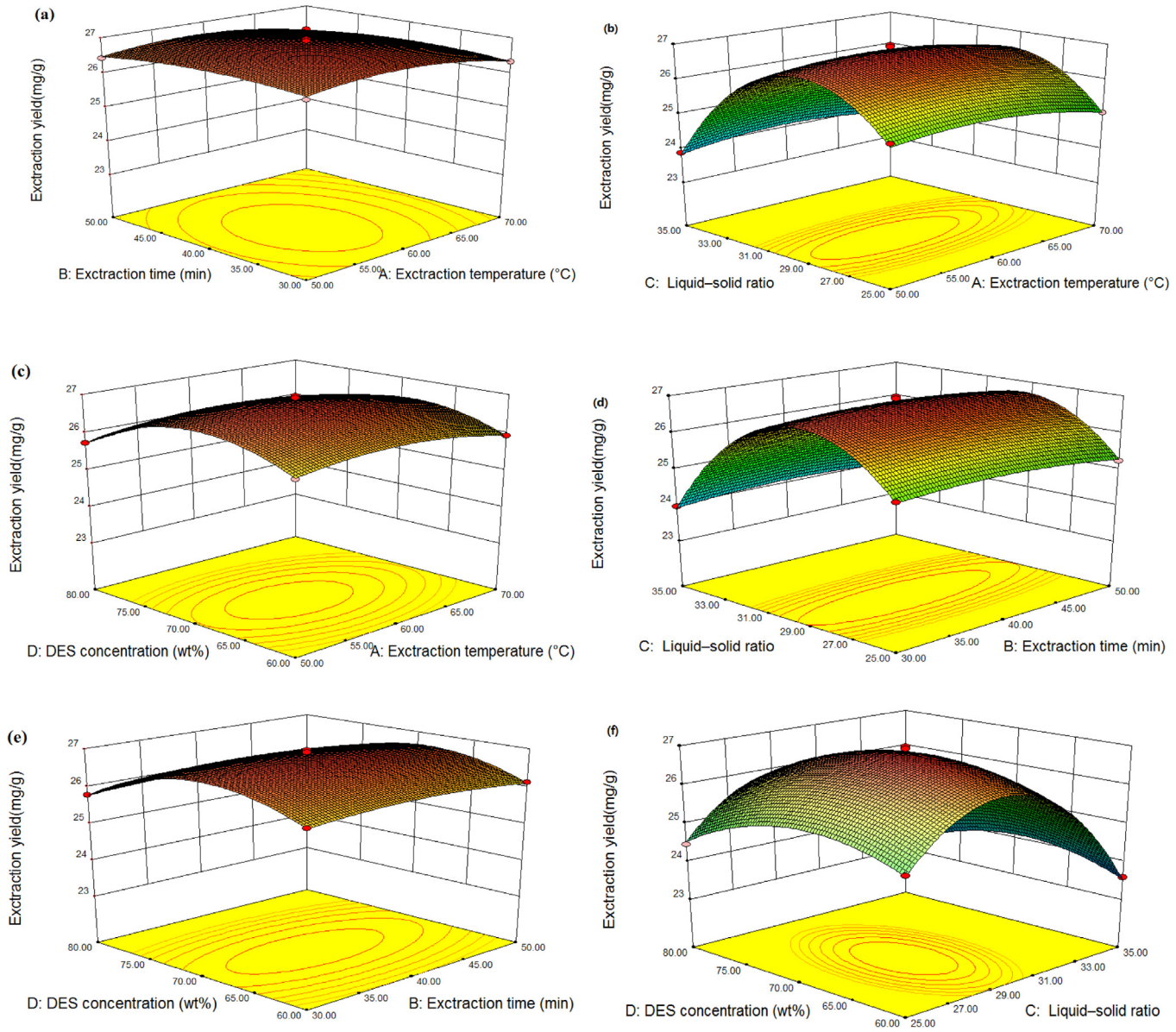


Fig. 3. Response surface plots for the extraction of caffeine reflecting the interaction between different factors. **(a)** extraction temperature and extraction time, **(b)** extraction temperature and liquid-solid ratio; **(c)** extraction temperature and DES concentration; **(d)** extraction time and liquid-solid ratio; **(e)** extraction time and DES concentration; and **(f)** liquid-solid ratio and DES concentration.

3.2.3. Effect of extraction time

The effect of extraction time of 10–50 min was studied. As shown in Fig. 2(c), the extraction yields significantly increased with the increase of extraction time from 10 to 30 min, then varied little when the extraction time is longer than 30 min. The increasing of time can make the CDT completely release into the extraction media. However, the longer the extraction time, the more energy will be consumed and the higher the cost. Therefore, the extraction time of 30 min was chosen for further studies by considering both the extraction yield and energy cost. The similar results were observed in other studies about the extraction of bioactive compounds from plant resources using DESs as extractant [41,42].

3.2.4. Effect of extraction temperature

The extraction temperature of 30–70 °C was investigated for this solid-liquid extraction. The extraction yields were shown in Fig. 2(d), which increased with increasing of the extraction temperature. And the maximum extraction yield was obtained at 70 °C, but not obviously higher than that at 60 °C. Appropriate rise of temperature is conducive to the increase of extraction yield and reduction of the system viscosity, but high temperature will also increase energy consumption. Therefore, the extraction temperature of 60 °C was selected by considering both the extraction yield and energy cost.

3.2.5. Effect of pH

The system pH of 1.0–13.0 for different samples was adjusted by the phosphate buffer solution to investigate the effect of pH. It can be seen from Fig. 2(e) that extraction yields decreased with the increase of pH, and the lower pH was more suitable for the caffeine extraction. It was reported that caffeine was an alkaline compound with a pK_a value of 10.4 [43], but the acidic condition was beneficial for the extraction. It can be interpreted as that the caffeine is in the form of cation under acidic condition, and the ionized form of caffeine is more easily being extracted than that in the molecular form. The pH value of the optimal extraction system (70 wt% ChCl-LA aqueous solution) is 1.14, therefore, the extraction experiments can be finished without adjusting the pH.

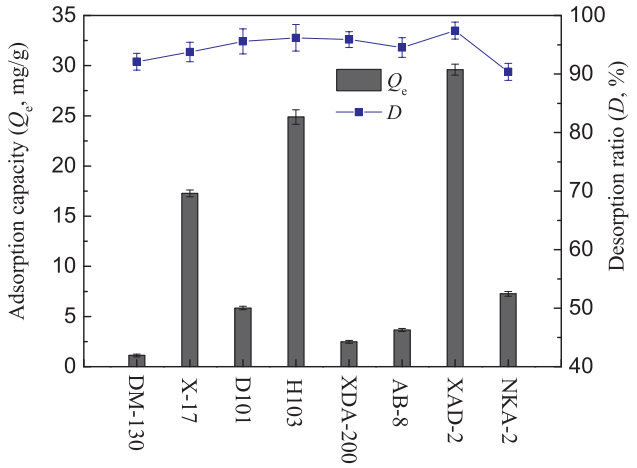


Fig. 4. Adsorption capacity (Q_e) and desorption ratio (D) for caffeine using different macroporous resins.

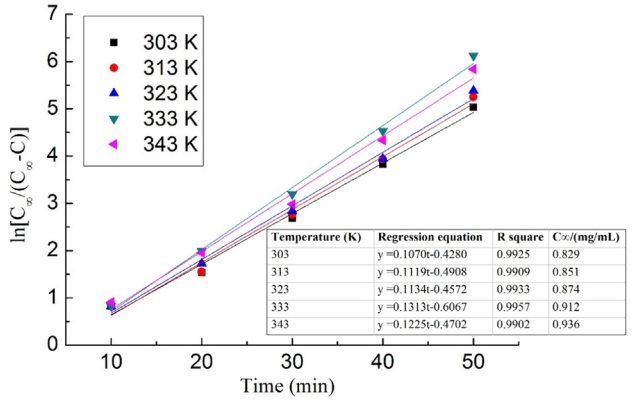


Fig. 5. Regression equations between ln[C_∞/(C_∞-C)] (y) and extraction time (t) at different temperature.

3.3. Optimization of the extraction by RSM

RSM based on BBD was developed to study the multiple factors and their mutual interactions, including extraction temperature, extraction time, liquid-solid ratio, and DES concentration. The experimental results for BBD were obtained in Table S2, and the analysis of variance (ANOVA) for the quadratic response surface model was shown Table 2. The ANOVA showed a P-value was < 0.0001, which indicated regression model was significant. The Model F-value of 542.84 implied that the model was significant. There is a < 0.01% probability that a "Model F-Value" occurs due to noise. The model terms can be considered to be significant when the P-values < 0.05. The model terms of *A, B, C, D, AC,* A^2, B^2, C^2, *and* D^2 were considered as significant terms according to this principle above. It can be seen from the F values that the factors influencing the caffeine extraction followed the orders: liquid-solid ratio > DES concentration > extraction temperature > extraction time. The "Lack of Fit F-value" was 0.61 and the "Lack of Fit P-value" > 0.05, which indicated that the model well explained all the data. The regression equation fitted to each factor by the response surface analysis was present in Eq. (11):

$$Y = 26.86 - 0.11 \times A - 0.045 \times B - 0.72 \times C - 0.22 \times D + 5.750 \times 10^{-4} \times A \times B + 0.085 \times A \times C - 0.025 \times A \times D + 0.048 \times B \times C - 0.013 \times B \times D + 0.049 \times C \times D - 0.29 \times A^2 - 0.17 \times B^2 - 2.02 \times C^2 - 0.75 \times D^2 (r^2 = 0.9982) \quad (11)$$

where Y represents the extraction yield (mg/g), A is the extraction temperature (°C), B is the extraction time (min), C is the liquid-solid ratio, and D is the DES concentration (wt%). The coefficient of determination (R^2) was 0.9982, implying that this quadratic model was quite suitable for the description of the response of the experiment. The response plots for the effect of the independent variables on the extraction yield (the mean for triplicate runs) of caffeine were shown in Fig. 3(a–f). It is well known that the steeper the curve in the response surface, the more obvious influence of one factor. So it can be seen from Fig. 3(a–f) that the liquid-solid ratio has the largest effect, and the extraction time has the smallest effect, which is consistent with the conclusion of the regression analysis in Table 2.

The design conditions of the caffeine extraction process were optimized by the software. The optimum conditions were as follows: extraction temperature of 57.92 °C, extraction time of 38.48 min, liquid-solid ratio of 29.07:1, and DES concentration of 68.68 wt%. The predicted value of extraction yield is 26.9552 mg/g under these conditions. The experiments were done at the optimized conditions (58 °C extraction temperature, 38 min extraction time, 29:1 liquid-solid ratio, and 69 wt% DES concentration) in triplicate runs, and the average extraction yield was 26.78 mg/g. The experimental result is much closer to the predicted value, which indicates this model is practical for the optimization of caffeine extraction using DESs as extractant.

3.4. Recovery of caffeine from DES extract by macroporous resin

The adsorption chromatography using macroporous resins was the most common method used for recovering the target compounds from the DES extract due to its simplicity and effectiveness [26,44–46]. Thus, eight kinds of macroporous resins were used to remove the DESs and other impurities from the crude extract. It can be seen from Fig. 4 that the largest Q_e (29.06 mg/g) and D (97.42%) values are obtained using XDA-200, which indicated that the XDA-200 had the good compatibility with caffeine. After removing the solvent, the dried sample was obtained, and the purity of caffeine reached 23.52% after enrichment and recovery. The resin was reused for two more times, the results for the second time were Q_e = 25.19 mg/g and D = 96.27%, and the results for the third time were Q_e = 23.73 mg/g and D = 95.12%.

3.5. Extraction kinetic study

The extraction yields for caffeine were obtained at different extraction (10–60 min) and temperatures (303–343 K) were obtained at the Table S3. It can be seen that more caffeine can be extracted with the increasing of extraction temperature at the same extraction time, which indicated that the high temperature is beneficial for the extraction. Simultaneously, more caffeine can be extracted with the increasing of extraction time (10–50 min) at the same extraction temperature, then the extraction yields varied little at the extraction time of 60 min, which indicated that the extraction tended to be balanced. The results of plotting the ln[C_∞/(C_∞-C)] versus the extraction time t were shown in Fig. 5. It can be observed in Fig. 5 that the linear relationship between ln[C_∞/(C_∞ − C)]and t is appropriate (R^2 > 0.99). The increase of temperature can accelerate the initial extraction rate and the enhance equilibrium extract concentration (the values of C_∞ slightly increased from 0.829 to 0.936), but it can't significantly improve extraction yield as the previous report [33]. This result was consistent with the conclusion in the single factor experiments and RSM optimization that temperature had no significant effect to extraction yield at the temperature range of 303–343 K. These results could provide a theoretical basis for industrial application and further studies of caffeine extraction from CDT.

4. Conclusions

DESs were developed for the extraction of caffeine for CDT leaves in this work. The DES ChCl-LA was screened as the optimal extractant. The

extraction conditions in the single factor experiments for preparing the crude extract were studied, then the main factors influencing the extraction were optimized by RSM. The extraction yield was 26.78 mg/g under the following optimized conditions: 58 °C extraction temperature, 38 min extraction time, 29:1 liquid-solid ratio, and 69 wt% DES concentration. Caffeine was enriched using XAD-200 macroporous resin to obtain a sample purity of 23.52%. The kinetics model for the extraction was also studied by the Fick's Law, this model was well fitted and can provide some guidance for the practical production. This extraction of caffeine using DES was much simpler, effective, and environmentally friendly, which has the potentiality in the scale-up production.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.seppur.2019.115723.

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